

ROS2

DAY 6 – Manual 2

Parameters – Implementation

Goal:

Create a ROS 2 Python node that:

- Has one parameter → `number_param`
 - Publishes that number to a topic `/number`
 - Allows changing the parameter at runtime using the terminal
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1. Start Fresh – Create a New Workspace

Step 1 — Open a new terminal

```
mkdir -p DAY-06/ros2_ws/src
cd DAY-06/ros2_ws
colcon build
source install/setup.bash
```

2. Create a ROS 2 Python Package

Step 2 — In the same terminal

```
cd src
ros2 pkg create --build-type ament_python param_pkg
```

This creates a package with:

- `package.xml`
 - `setup.py`
 - `param_pkg/`
-

3. Create the Parameter + Publisher Node

Step 3 — Open the Python file

```
cd param_pkg/param_pkg
nano number_node.py
```

Paste the code below

```
import rclpy
from rclpy.node import Node
from std_msgs.msg import Int32

class NumberParamNode(Node):
    def __init__(self):
        super().__init__('number_param_node')

        # Declare parameter
        self.declare_parameter('number_param', 10)

        # Publisher
        self.publisher = self.create_publisher(Int32, 'number', 10)

        # Timer
        self.timer = self.create_timer(1.0, self.timer_callback)

        self.get_logger().info("Node started. Publishing number_param every second.")

    def timer_callback(self):
        number_value = self.get_parameter('number_param').value
        msg = Int32()
        msg.data = number_value
        self.publisher.publish(msg)
        self.get_logger().info(f"Published: {number_value}")

def main(args=None):
    rclpy.init(args=args)
    node = NumberParamNode()
    rclpy.spin(node)
    node.destroy_node()
    rclpy.shutdown()

if __name__ == '__main__':
    main()
```

Save and exit:
Press Ctrl + O, Enter, then Ctrl + X.

4. Make Executable Entry Point

Step 4 — Edit setup.py

```
cd ../  
nano setup.py
```

Find:

```
entry_points={  
    'console_scripts': [  
    ],  
},
```

Replace with:

```
entry_points={  
    'console_scripts': [  
        'param_node = param_pkg.param_node:main'  
    ],  
},
```

Save and exit.

5. Build the Workspace

Step 5 — Open a new terminal

```
cd DAY-06/ros2_ws  
colcon build  
source install/setup.bash
```

6. Run the Node

Step 6 — Open a new terminal (Terminal 1)

Run the node:

```
ros2 run param_pkg param_node
```

Expected output:

```
Node started. Publishing number_param every second.  
Published: 10  
Published: 10  
Published: 10  
...
```

7. View the Topic Data

Step 7 — Open another terminal (Terminal 2)

```
ros2 topic echo /number
```

Output:

```
data: 10  
data: 10  
data: 10  
...
```

8. Change the Parameter in Real-Time

Step 8 — Open another terminal (Terminal 3)

Change parameter:

```
ros2 param set /number_param_node number_param 99
```

Terminal 1 output instantly updates:

```
Published: 99  
Published: 99  
...
```

Terminal 2 output:

```
data: 99  
data: 99  
...
```

9. Run Node With a Different Value (Override at Start)

Step 9 — Terminal 1

Stop node (Ctrl + C) and run:

```
ros2 run param_pkg param_node --ros-args -p number_param:=42
```

Output:

```
Published: 42  
Published: 42  
...
```

10. Summary

We have learned:

- Creating workspace
 - Creating ROS 2 package
 - Writing a node with one parameter
 - Publishing parameter to a topic
 - Changing parameter during runtime
 - Viewing topic echo
 - Overriding parameters from terminal
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